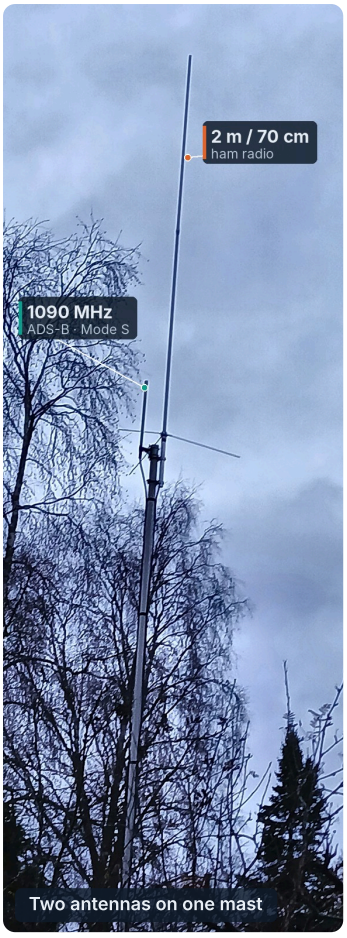
MODE-S Wind · the receiving station

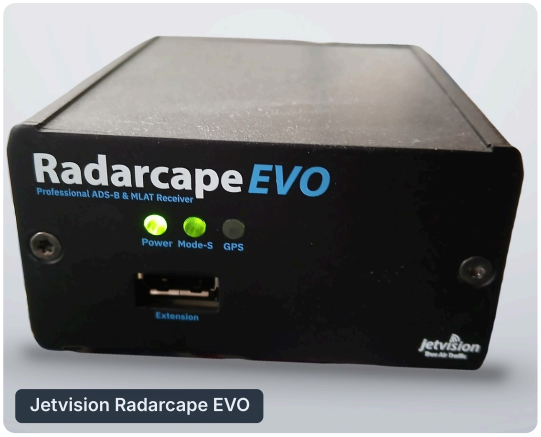
Airliners as weather sensors — wind, temperature and the heights of the pressure surfaces from the Mode S and ADS-B replies heard in Tuusula, north of Helsinki-Vantaa (EFHK)

Otso Laakso · OH2GAX
Version v.27102026
October 2026 · 1 / 4

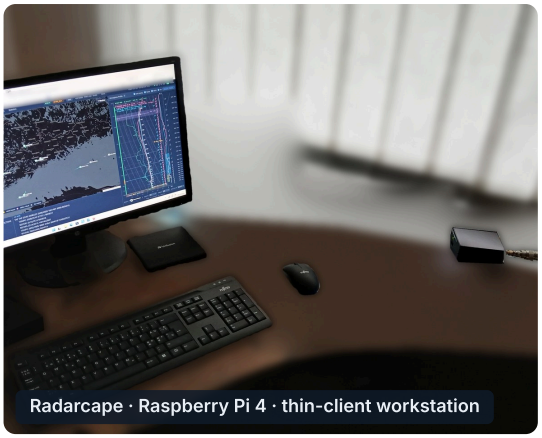
1 HARDWARE — one ADS-B antenna, one receiver, one small computer; the users are served from the cloud (page 4)



Two antennas on one mast



Jetvision Radarcape EVO


Radarcape · Raspberry Pi 4 · thin-client workstation

Signal chain

Aircraft transponder › 1090 MHz › GP antenna › coax › Radarcape › Ethernet › Raspberry Pi 4 › browser · phone

Antenna
Two antennas share the mast: the **short ground-plane without radials** is the 1090 MHz ADS-B / Mode S antenna, the tall one with radials the author's 2 m / 70 cm ham-radio antenna. Aircraft at cruise levels are heard to about 160 NM.

Raspberry Pi 4
The local station: the Python collector with **pyModeS** decoding, the SQLite database, the hourly jobs and the web server — and the encrypted tunnel that feeds the cloud server.

Jetvision Radarcape EVO
An FPGA Mode S decoder with GPS-disciplined nanosecond timestamps — the basis of multilateration (MLAT). Every frame goes out as **Beast binary** over TCP (with its signal level), the aircraft list with MLAT positions as JSON.

Thin client and phone
Viewers: the pages run in any browser, and /app gives the Skew-T, time–height, ILS approach and approach conditions on a phone. The i5 thin client becomes the next local station.

2 WHAT THE AIRCRAFT SEND

The ground radars and Fintraffic's **wide-area multilateration (WAM)** network interrogate the aircraft on 1030 MHz, and the transponders answer on **1090 MHz**. The station only **listens**: it hears the answers, never the questions.

In Europe, Mode S **Enhanced Surveillance (EHS)** makes the larger aircraft answer with **Comm-B registers** — 56-bit snapshots of their own air-data and inertial systems: true airspeed, Mach, magnetic heading, roll, track and ground speed. Air traffic control uses them to see what the aircraft is doing; MODE-S Wind uses the same numbers to measure the air the aircraft flies through. On top come the aircraft's own **ADS-B** squitters with GNSS positions and the GNSS – barometric height difference.

Because the request is not heard, the reply does not say which register it holds. pyModeS infers it from the content (and finds the aircraft's address hidden in the reply's parity bits); ambiguous replies are dropped, and the Statistics page counts how often this happens.

≈ 130 /s
Mode S and ADS-B messages, 24-hour mean

1.3–1.9 s
median interval between one aircraft's BDS 5,0 replies (en route → final)

945 000
stored observations, 24 Sep – 26 Oct 2026

≈ 85 %
of the observations with a wind pair also give a temperature

FORMATS AND REGISTERS — used by MODE-S Wind

Format · register	Content
`DF17 ADS-B`	Sent by the aircraft itself: positions (TC 9–18), velocity with the **GNSS – baro height difference** (TC 19, about twice a second), operational status; NACp / NIC give the GNSS quality
`DF20 / DF21`	Comm-B reply to a radar or WAM request, with altitude (20) or squawk (21); carries one 56-bit register:
`BDS 5,0`	**Track and turn report**: roll angle, true track, ground speed, track-angle rate, true airspeed
`BDS 6,0`	**Heading and speed report**: magnetic heading, indicated airspeed, Mach, barometric and inertial vertical rate
`BDS 4,0`	Selected vertical intention: selected altitude and the pilot's QNH setting
`BDS 4,4 / 4,5`	Meteorological routine (MRAR) and hazard (MHR) reports — rare here; MHR temperatures are rejected (mostly other registers misidentified)
`BDS 1,0 · 1,7 · 2,0`	Data-link capability · supported registers · callsign
`DF11`	All-call reply — its parity carries the **interrogator code** (II / SI): which radar or WAM station is asking
`DF4 / DF5`	Surveillance replies: altitude, squawk — the reply rhythm per flight phase
`DF0 / DF16`	Airborne collision avoidance (ACAS / TCAS) traffic

Quality gates — GNSS values only with NACp ≥ 8 and outside the airframe's jamming episodes (from the GPS Quality monitor); MLAT keeps aircraft placed when jamming removes their ADS-B positions; WAM ground stations, helicopters and frozen or spoofed positions are filtered out before any meteorological use.

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Decoding with pyModeS (Junzi Sun, TU Delft) · method after Siebren de Haan (KNMI)

3 FROM REPLIES TO WEATHER

Wind — ground vector minus air vector

```
W = (GS, true track) − (TAS, true heading)
true heading = magnetic heading
  + declination − δ(airframe)
```

A BDS 5,0 and a BDS 6,0 reply of the same aircraft at most 10 s apart; none while it banks or turns fast, and on final only from replies at most 6 s old. The declination comes from the World Magnetic Model 2025 at the aircraft's own position (8.7–12.6° around Helsinki) — 1° of heading is about 8 kt of wind at 450 kt.

Temperature — from the air-data computer

$a = TAS / M, \quad T = a^2 / (\gamma R)$

$T\,[K] = 288.15 \cdot (TAS / (661.47\ kt \cdot M))^2$

The static air temperature follows from the true airspeed (2 kt steps) and the Mach number (0.004 steps) of the same aircraft: about ±1.3 K per reply at cruise and ±3–4 K on approach, so profiles use trimmed means of many replies and aircraft.

Heights of the pressure surfaces

$z = (p\text{-alt} + \Delta_{GNSS-baro}) \cdot 0.3048 - N_{EGM96} - curve(alt) - offset$

$T_v = g_0\ \Delta H / (R \ln(p_1/p_2))$

The pressure altitude tells which pressure surface the aircraft flies on, the GNSS difference how high that surface lies: D-values, height contours and the geostrophic wind, and from every climb an independent layer temperature.

4 SELF-CORRECTION & CALIBRATION — learnt from the data itself, checked against independent references

Aircraft hundreds of airframes a day, each with its own small instrument errors ⇄ **Each other** every airframe is compared with the other aircraft of the same layer and hour ⇄ **Jokioinen radiosonde** FMI open data, soundings at 00, 06 and 12 UTC: the independent truth ⇄ **MEPS model** MET Norway open data, every hour, taken at the aircraft's own positions

Heading offset of each airframe

Many avionics convert their inertial true heading to magnetic heading with a magnetic-variation table that is years old, while the declination at Helsinki has grown from 7.0° (2005) to 10.5° (2026). A heading offset shows as a wind error across the heading, proportional to the airspeed, so each airframe's offset δ is learnt from its wind against the other aircraft of its layer (fleet mean +1.5°, E190 3.5°, B777 ≈ 0°), faded with a 30-day half-life; by type or fleet median until the airframe has two flights.

Wind against the soundings 11.5 → 9.1 kt RMS; single-aircraft scatter halved.

Upper-level temperature

The fleet's Mach/TAS temperature reads about +0.7 K warm at 300–250 hPa — against both the model and the soundings, steady for a month. A correction per level (400–200 hPa) is learnt every hour from the MEPS check of the last 21 days and added when the Skew-T and time–height are served, faded out below FL180. A per-airframe temperature calibration was tested and rejected: it gained only 1 %.

200–300 hPa error against the soundings 1.27 → 0.96 K RMS.

GNSS heights of the pressure surfaces

After every sounding the aircraft that passed a level within ±1 h and 150 km of the drifting balloon are compared with its measured height of that surface. The fit gives a curve per 5 000 ft band — the altimeter's speed-dependent static-pressure error — and a small, shrunk offset per airframe, refitted from 30 days. The EGM96 geoid (about 18 m here, sloping 3 m per 100 km) is removed at every aircraft position.

Within ±2 m of MEPS at 400–250 hPa, hourly correlation 0.996–0.999.

Airspeed, declination and the checks

When an aircraft sends Mach but no TAS, the airspeed uses the live ISA deviation measured by the other aircraft per 5 000 ft band. Every sounding gets a **Sonde check** and every hour a **MEPS check**, with the model taken where the aircraft actually flew. Raw values are always stored and the corrections are applied when served, so they can be learnt again and checked independently — and everything works without the model.

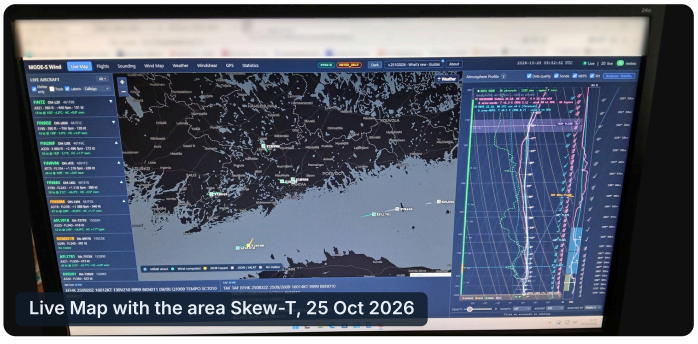
Area wind within 2.4–3.1 kt (vector RMS) of MEPS at 925–400 hPa.

5 DATA STORAGE — SQLite (WAL) on each server

Table / data	Content	Kept
`observations · flights`	per-aircraft rows: wind and temperature (raw and corrected), GNSS difference, sources	configurable
`profile_hours`	hourly area profile archive (Skew-T, time–height)	indefinitely
`approach_history`	every landing: wind per 200 ft, ACI, windshear report	indefinitely
`gnss_hours · height_series`	hourly height cells · dense climb / descent series	long · 30 d
`heading_cal · t_correction · gnss_cal`	the learnt calibrations, saved hourly	current
`meps_* · soundings/`	model profiles (kept) and maps · radiosondes as files	maps 7 d · 1 year
`gps_quality_* · gps_episodes`	GNSS degradation by hour and episode	manual purge
`msg_stats_hours`	receiver statistics, one row an hour	7 d

Light on the SD card by design. The live aircraft state, the windshear and GPS trackers and the message statistics live in RAM; a per-aircraft write throttle and an optional meteo-only mode limit the rows; one batched writer thread and hourly jobs that write one row per hour. About 12 MB of database a day; the Maintenance page shows every table and purges by date.

6 WHAT THE SYSTEM SHOWS


Live Map with the area Skew-T, 25 Oct 2026

Live Map aircraft by meteo source, trails, wind barbs, MEPS or radar weather, live area Skew-T

Sounding Skew-T now, any hour or per flight; time–height; stability indices; sonde and MEPS checks

Wind Map wind and temperature at a level, isotachs, streamlines, heights, geostrophic wind, cross-sections

Windshear ILS profile of every approach, headwind-ramp alerts (ADV / WSA / MBA), go-arounds, ACI — page 3

Weather · GPS MEPS maps with a 48 h forecast · area-wide GNSS jamming monitor and episodes

Statistics · /app formats, registers, interrogators, range rings, system health · the phone view

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7 WINDSHEAR DETECTION — from the aircraft's own replies, for every landing and departure

Low-level windshear is a sudden change of the wind along the approach or climb-out path. ICAO counts it below 1 600 ft above the runway; a headwind change of 15 kt or more, lasting longer than a gust, is significant. What matters to an aircraft is the change of the **headwind along its own path** — and that is exactly what each aircraft reports in its **BDS 5,0** replies: the difference of its true airspeed and ground speed.

The station follows this headwind about every two seconds along the path of every aircraft on final and on the climb-out (noise about 1.1 kt per reply, a new value about every 290 m). The method follows the **Hong Kong Observatory**, which finds headwind **ramps** along the glide path in Doppler-lidar scans — the lidar's one-dimensional headwind profile is the same thing an approaching aircraft samples itself.

`headwind = TAS − GS (each BDS 5,0 reply)`

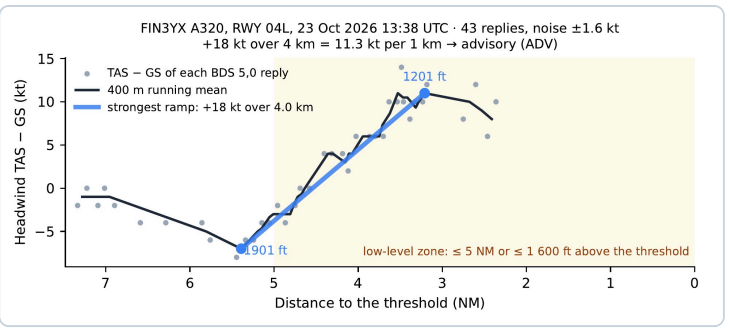
$\Delta V_{1\ km} = \Delta V \cdot (1\ km\ /\ L)^{1/3}, \quad L = 400 \ldots 4\,000\ m$

Replies in a bank, repeated registers and spikes are dropped; the profile is a 400 m running mean along the path. The strongest loss and gain of each flight are scaled to a 1-km equivalent, so a quick change counts more than a slow one. Only the zone below 1 600 ft above the threshold or within 5 NM counts.

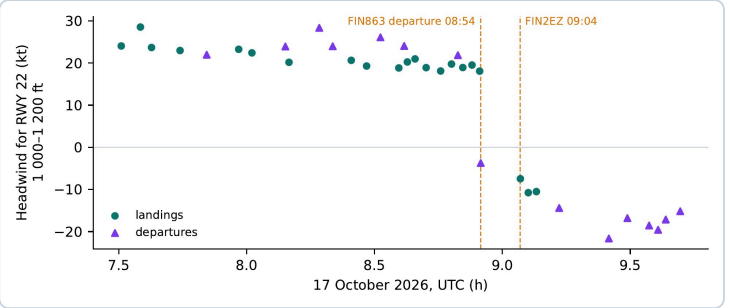
ΔH_{band} `= headwind(200-ft band) − median of the flights of the last 45 min`

A front or gust front shows as a change *between* flights: each 200-ft band from 400 to 1 800 ft is compared, on the wind vector, with the landings and departures of the previous 45 minutes — landings with landings and departures with departures, because the airspeed reads about 1.4 kt differently in the two flight phases.

Weak ramps (10–12 kt) count only when they stand out from what the other flights met at the same heights; each ramp alert is marked **confirmed** when another landing within 15 minutes saw the same change at the same height, or **isolated** (probably turbulence) when none did. The detection runs on the Pi for every flight, whether or not a page is open; every landing and departure gets a stored **windshear report**, and the Windshear page and `/app` show live alerts, a 24-hour log and the ILS profile.



A headwind ramp. The headwind of one landing, reply by reply (dots) and as the 400 m running mean, with its strongest ramp — a persistent shear layer before the front of 23 Oct 2026 (advisory).



A change between flights, 17 Oct 2026. Headwind at 1 000–1 200 ft of the landings and departures on runway 22: a departure met 25 kt less headwind ten minutes before the first landing; three landings in a row then confirmed it (band WSA), before the wind reversed and runway 04 came into use.

ADV advisory ≥ 10 kt **WSA** alert ≥ 15 kt **MBA** microburst ≥ 30 kt loss

8 THE ONGOING STUDY — windshear from passively received Mode S EHS at Helsinki-Vantaa

No published study has detected low-level windshear from Mode S EHS replies received passively. MODE-S Wind tests it at EFHK, and every step is documented with its reason and checked on the stored flights — a **study log** for aviation meteorologists and atmospheric scientists, updated as the study goes on (`doc/EFHK_MODE_S_EHS_Windshear_Detection_Study.pdf`; how to read the page: `doc/Windshear_User_Guide.pdf`).

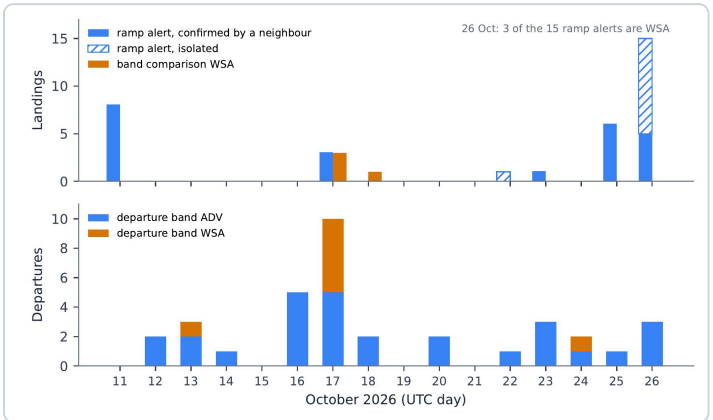
May – 24 Oct 2026
Six browser algorithms
A replay of 3 084 landings in calm weather: they mostly reacted to the normal increase of the wind with height — retired.

24 Oct
Ramps and band comparison
The Hong Kong method on the server; bands on the wind vector with departures as references.

26 Oct
Weak advisories relative
A stable evening flagged 10 of 39 landings for the expected profile; advisories 59 → 26.

27 Oct
Confirmed or isolated
A turbulent southerly day: one real shear layer seen by four landings among single-aircraft eddies.

27 Oct
By flight phase
Departures read +1.4, landings −1.5 kt against the MEPS model — an airspeed effect, now removed.



Windshear flags per day, 11–26 Oct 2026 (current method on all stored flights). Top: landings — ramp alerts confirmed or isolated, and band-comparison WSAs. Bottom: departures.

3 778 · 3 539
landings · departures with a windshear report, 9–27 Oct 2026

36 · 25 ✔
landings with a ramp alert · of them confirmed by a neighbour (11 isolated)

0.9 % · 0.08 %
landings with an advisory · with a windshear alert from the ramps

232–733 ft
lowest height followed above the threshold, by runway

Next

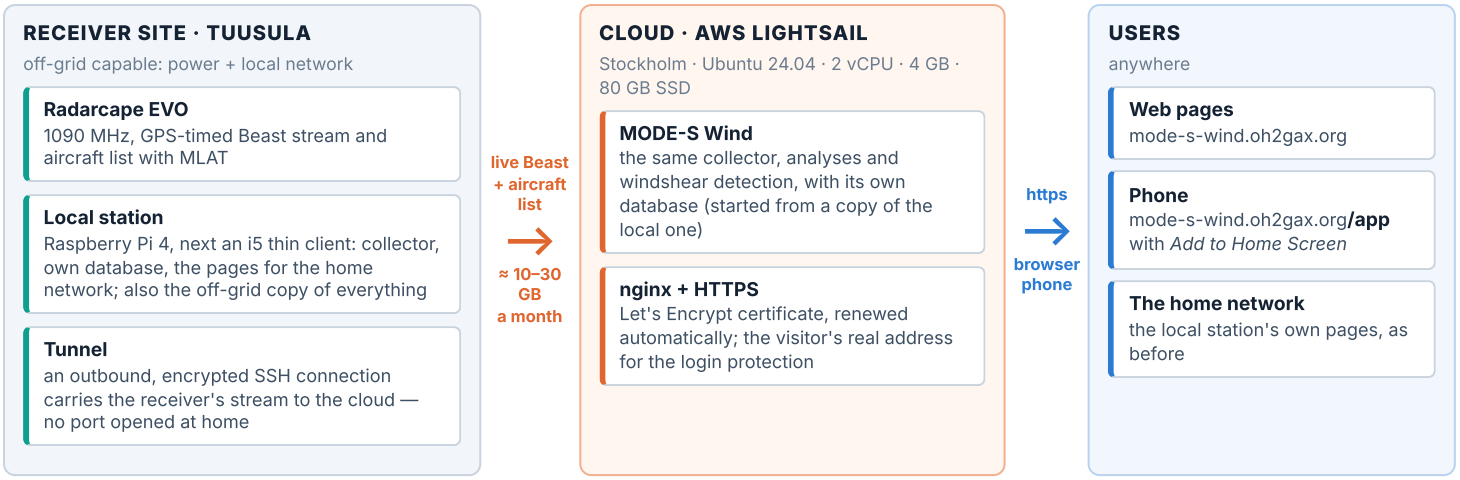
- **ADS-B ground speed** (about twice a second) to sharpen the ramps — test from early November 2026
- **Convective season 2027**: gust fronts and microburst outflows are the real test of the levels
- **Validation** against METAR / SPECI windshear reports and go-arounds; an EFHK low-level wind climatology

Research only — passive and uncertified; the alerts describe air already flown through. Not for flight operations or ATC.

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9 ONE SYSTEM, TWO SITES — from 27 October 2026

MODE-S Wind now runs in two places with the **same software**. The **local station** at the receiver site keeps the original idea: with power and the local network alone, the receiver's data flows into its own database — no internet needed (only the model, sounding and METAR downloads wait for it). A **cloud server** at Amazon Web Services runs a complete second copy of MODE-S Wind, fed live from the same receiver, and serves all users at **https://mode-s-wind.oh2gax.org**.



One code base on GitHub (oh2gax/mode_s_wind) — each server updates itself with `bash update.sh`, the local station first and the cloud when it runs well; the few settings in which they differ are kept on each server in `config_local.py`.

	Local station	Cloud server
Role	development station; off-grid observing site	production: the public service for all users
Network	home fibre; the home network	data-centre network, 4 TB of traffic a month included
Without internet	keeps collecting, analysing and serving the home network	no new data until the tunnel is back
Database	SQLite on the station; the reference of the data checks and the windshear study	SQLite on SSD; backed up before every update
Calibrations	learnt independently on each server (heading offsets, upper-level temperature, GNSS heights) — they stay close, and each can be checked against the other	

2 copies
of the database, each complete — at the receiver site and in the cloud

1 tunnel
outbound and encrypted: the cloud sees the receiver as if it were next to it

≈ 10–30 GB
of Beast data a month to the cloud, plus the aircraft list every 2 s

1 command
updates a server from GitHub, with a database backup and a way back

10 WHY A CLOUD SERVER

- **Room for many users.** Every open page asks for new data every few seconds and the Live Map keeps a live stream open. A Raspberry Pi on a home connection serves a handful of users; the cloud server has a data-centre connection and its own processor, memory and SSD for them.
- **The receiver site is not exposed.** Users connect to the cloud, never to the home network; the site's only connection is outbound.
- **Secure and simple to reach.** A permanent https address with a certificate, also for the phone view.
- **Two independent copies.** The local station collects even when the internet or the cloud is down; the cloud keeps serving when the station is serviced or replaced.

11 KEPT IN STEP

new commits › database backup › update (fast-forward only) › compile check › restart › log check

`update.sh` does every update the same way: it lists what is new, copies the database, takes GitHub's version, checks the changed program files — going back at once if one fails — restarts the service and shows its first log lines; `--rollback` returns to the previous version. Settings that differ between the servers live only on each server, so an update never overwrites them.

Development at home, production in the cloud. A change runs first on the local station, with the same live data, before it reaches the users. The local station's former users are pointed to the new address by a notice on its pages.

12 NEXT — growing from one receiver into a network

soon
i5 thin client
becomes the local station; the Raspberry Pi stays as a dedicated tunnel box, so the cloud's feed no longer depends on the local server.

planned
Filling gaps
after an outage, each server copies the missing hours from the other — two databases, nothing lost.

planned
Second receiver
close to EFHK, with a view of every threshold: approaches and departures followed down to the runway.

later
Own multilateration
three or more GPS-timed receivers: aircraft positions independent of GNSS jamming, also at low level.

Status, 27 October 2026. The cloud server went live with HTTPS on its first day, fed through the tunnel from Tuusula and started from the local station's database; both servers were updated from GitHub with `update.sh` the same evening, and the users of the local station are being guided to **https://mode-s-wind.oh2gax.org**.

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